Supply Chain & Courier Billing Analysis

A Deep Dive into Order Fulfillment, Shipping Costs & Billing Discrepancies for Company X

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Project Overview

Business Context

Company X is an e-commerce seller that ships products through a third-party courier service. Each shipment is billed based on weight, delivery zone, and shipment type.

This project analyzes the complete order and courier data to verify billing accuracy, understand shipping patterns, and recommend cost optimization strategies.

Audit

Verify courier billing against expected charges

Analyze

Understand zone-wise and weight-wise cost drivers

Optimize

Identify areas for shipping cost reduction

Report

Provide actionable insights for management

Dataset Description

Order Report 400 rows × 3 cols	Order IDs, SKUs, and order quantities. Contains 124 unique orders and 65 unique SKUs.
Courier Invoice 124 rows × 8 cols	AWB code, Order ID, charged weight, warehouse/customer pincode, zone, shipment type, billing amount.
SKU Master 66 rows × 2 cols	Maps each SKU to its actual weight in grams. Enables expected weight calculation.
Pincode Zones 124 rows × 3 cols	Maps warehouse–customer pincode pairs to delivery zones (a–e).
Courier Rates 1 row × 20 cols	Rate card: fixed + per-500g additional charges for each zone and direction (forward/RTO).

Key Metrics at a Glance

A snapshot of the complete dataset

124 Total Orders	**400** Line Items	**65** Unique SKUs
₹13,648 Total Billed (Rs.)	**₹110.07** Avg. Billing / Order	**12.1%** RTO Rate

Order Report Analysis

Key Findings

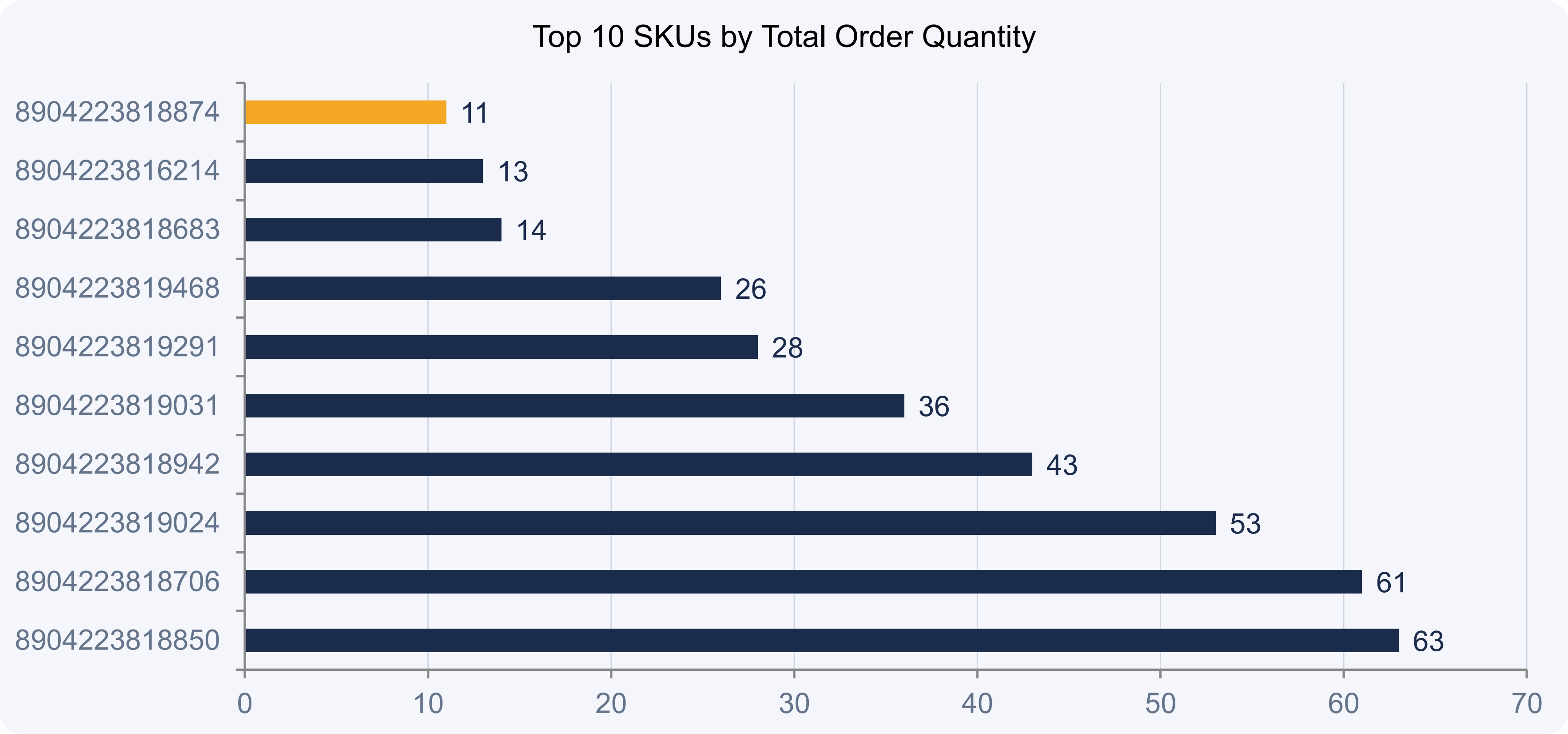
4.2 Average items per order

58 Orders with exactly 3 items

91.1% Orders with 2+ items

18 Highest items in one order

Top Selling SKUs



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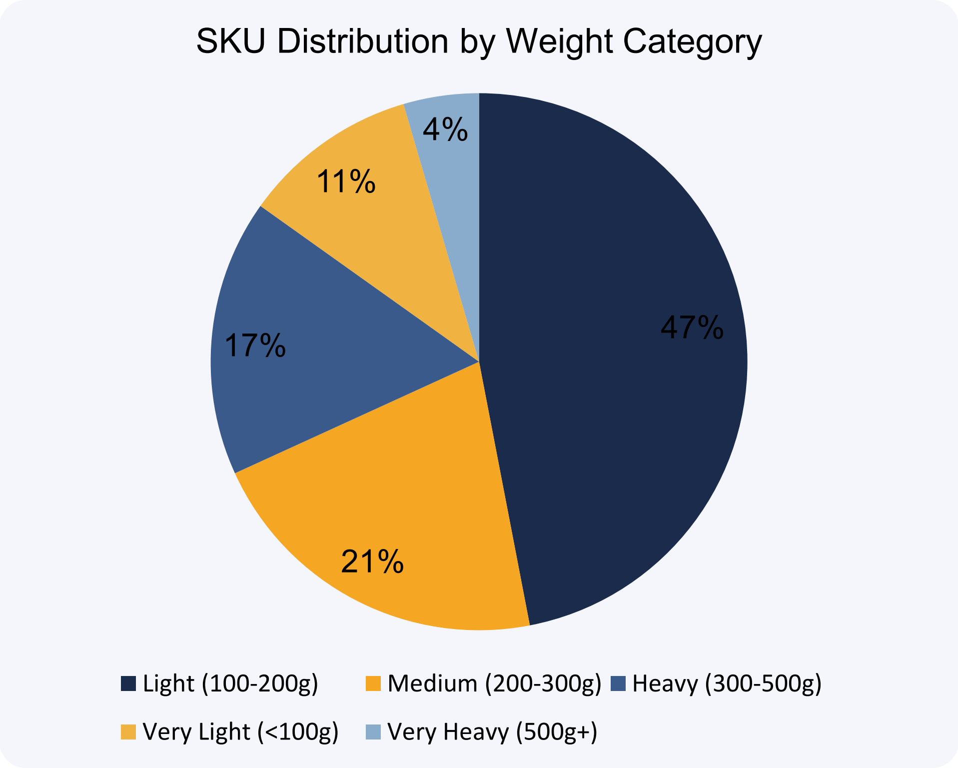
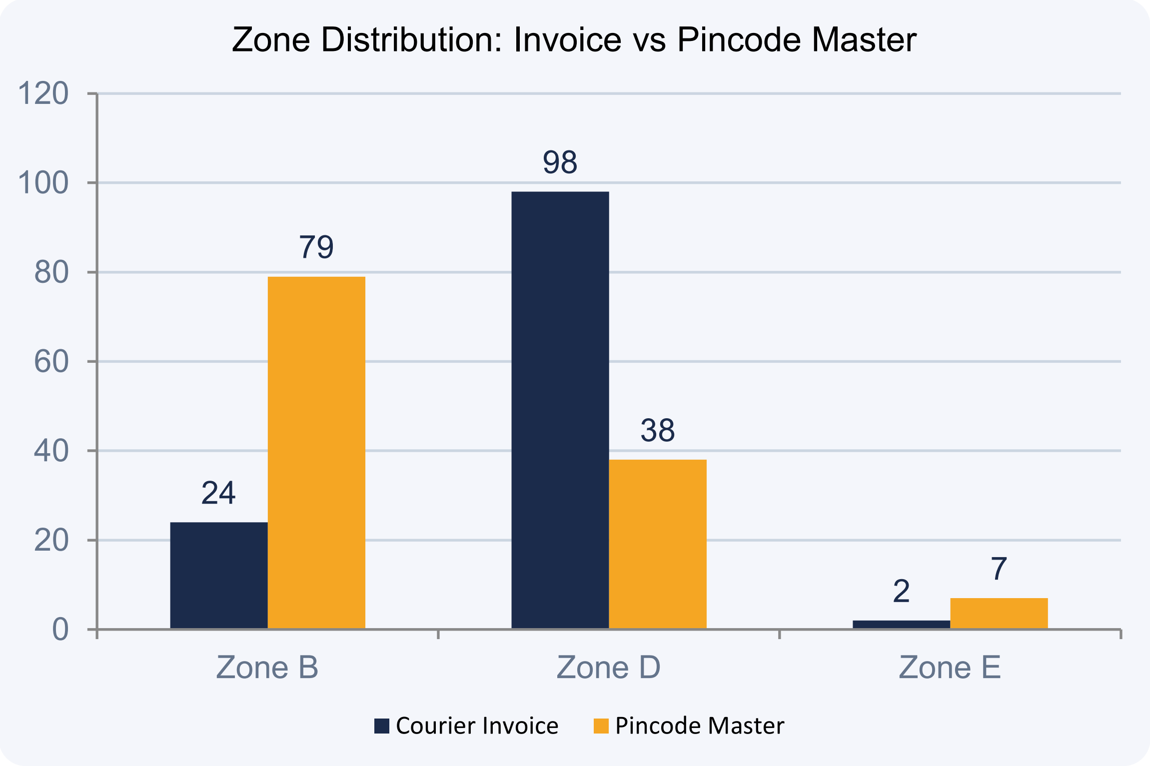
SKU Weight Analysis



Product Weight Profile

219.7g	Average SKU Weight
165g	Median SKU Weight
10g	Lightest SKU
600g	Heaviest SKU
66	Total Unique SKUs
47%	SKUs are Light (<200g)

Zone Distribution Analysis



Zone Mismatch Alert

A significant discrepancy exists between the courier's assigned zones and the pincode master zone data.

Zone D: Invoice shows 98 shipments, but pincode master shows only 38.

Zone B: Invoice shows 24, but pincode master shows 79.

This zone misclassification is a major billing risk and warrants immediate investigation.

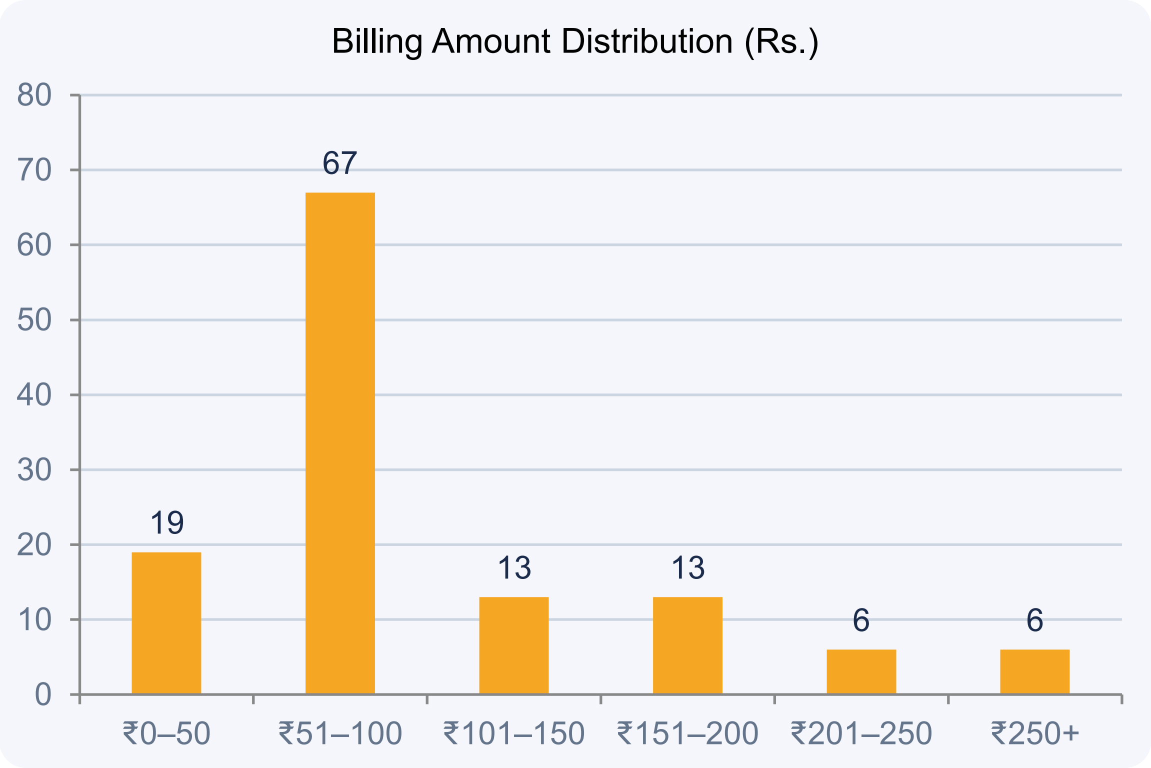
Zone Classification: Invoice vs Pincode Master

Zone	Invoice Count	Pincode Master Count	Difference	Impact
Zone B	24	79	-55	Underclassified — courier charging less
Zone D	98	38	**+60**	Overclassified — higher zone = higher cost
Zone E	2	7	-5	Minor underclassification

What Does This Mean?

Zone D has significantly higher rates than Zone B. If 60 shipments were classified as Zone D when they should have been Zone B, Company X may have been overcharged. Conversely, Zone B underclassification means potential revenue loss for the courier, but an advantage for Company X. This discrepancy needs reconciliation with the courier company.

Courier Invoice Overview



₹13,648 Total Amount Billed

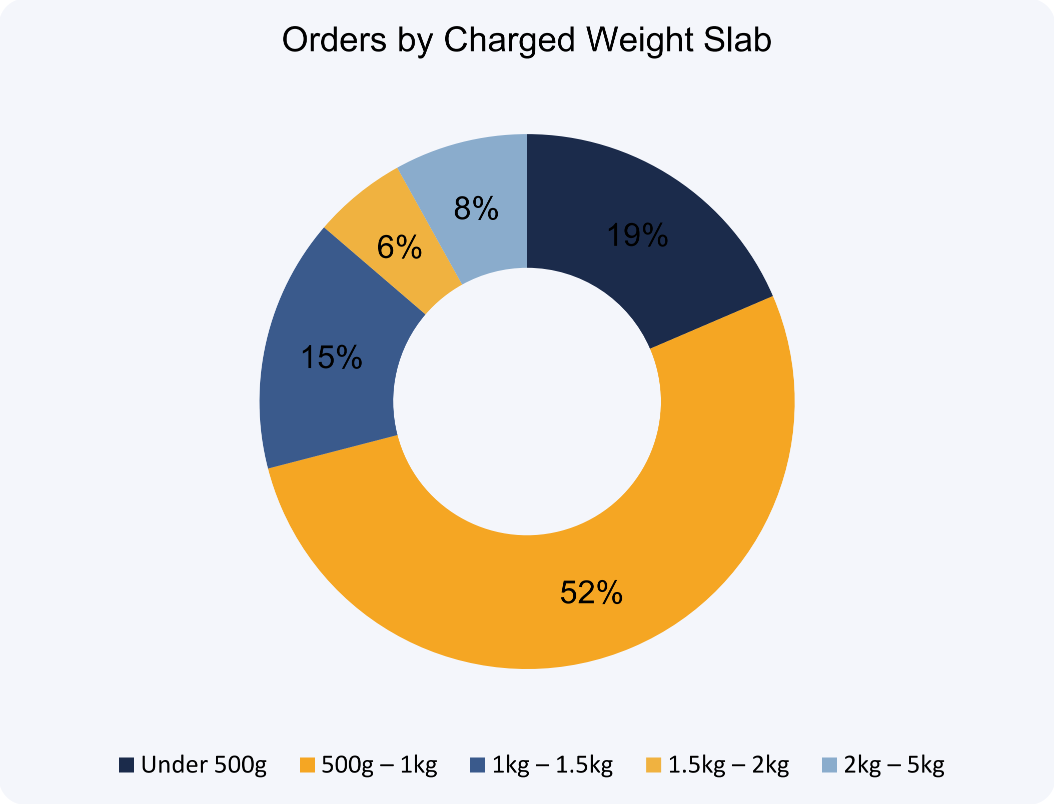
₹110.07 Average per Order

₹33 Minimum Charge

₹403.8 Maximum Charge

₹90.2 Median Charge

Charged Weight Distribution



Insights

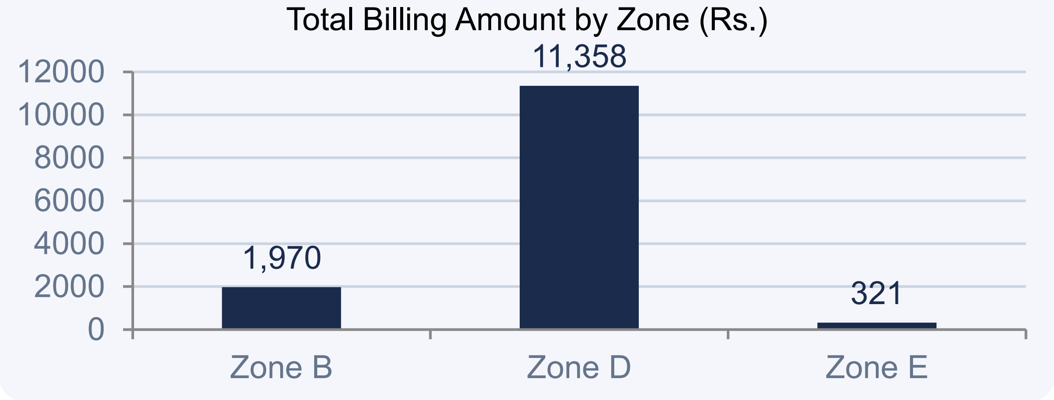
1. 52.4% of all shipments fall in the 500g–1kg bracket, which is the most common billing slab.

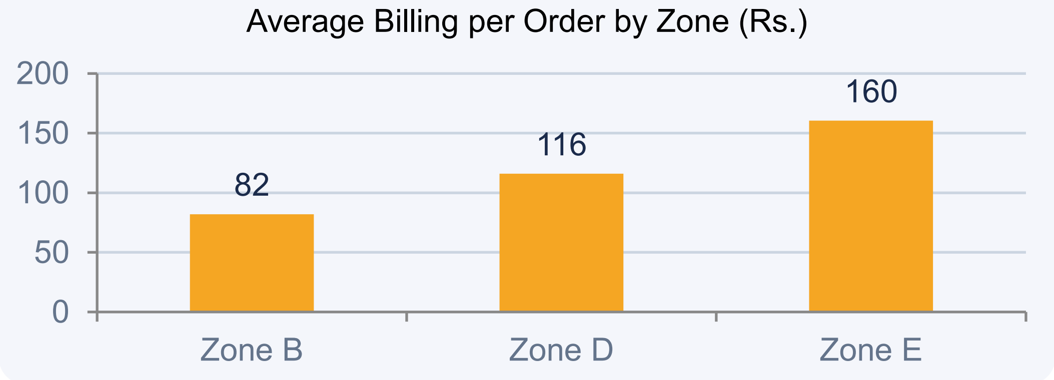
2. 18.5% are under 500g — potential for weight bundling to reduce per-order costs.

3. Only 2 shipments exceed 5kg, indicating Company X deals primarily in lightweight products.

4. Average charged weight is approximately 0.9 kg per shipment.

Billing Analysis by Zone





ZB

24 orders

₹1,969.9 total

₹82 avg

ZD

98 orders

₹11,357.5 total

₹115.9 avg

ZE

2 orders

₹320.8 total

₹160.4 avg

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Courier Rate Structure

Forward Charges (Rs.) — Fixed base + Additional per 500g slab

Zone	Fixed (Rs.)	Additional/500g (Rs.)
A	29.5	23.6
B	33.0	28.3
C	40.1	38.9
D	**45.4**	**44.8**
E	56.6	55.5

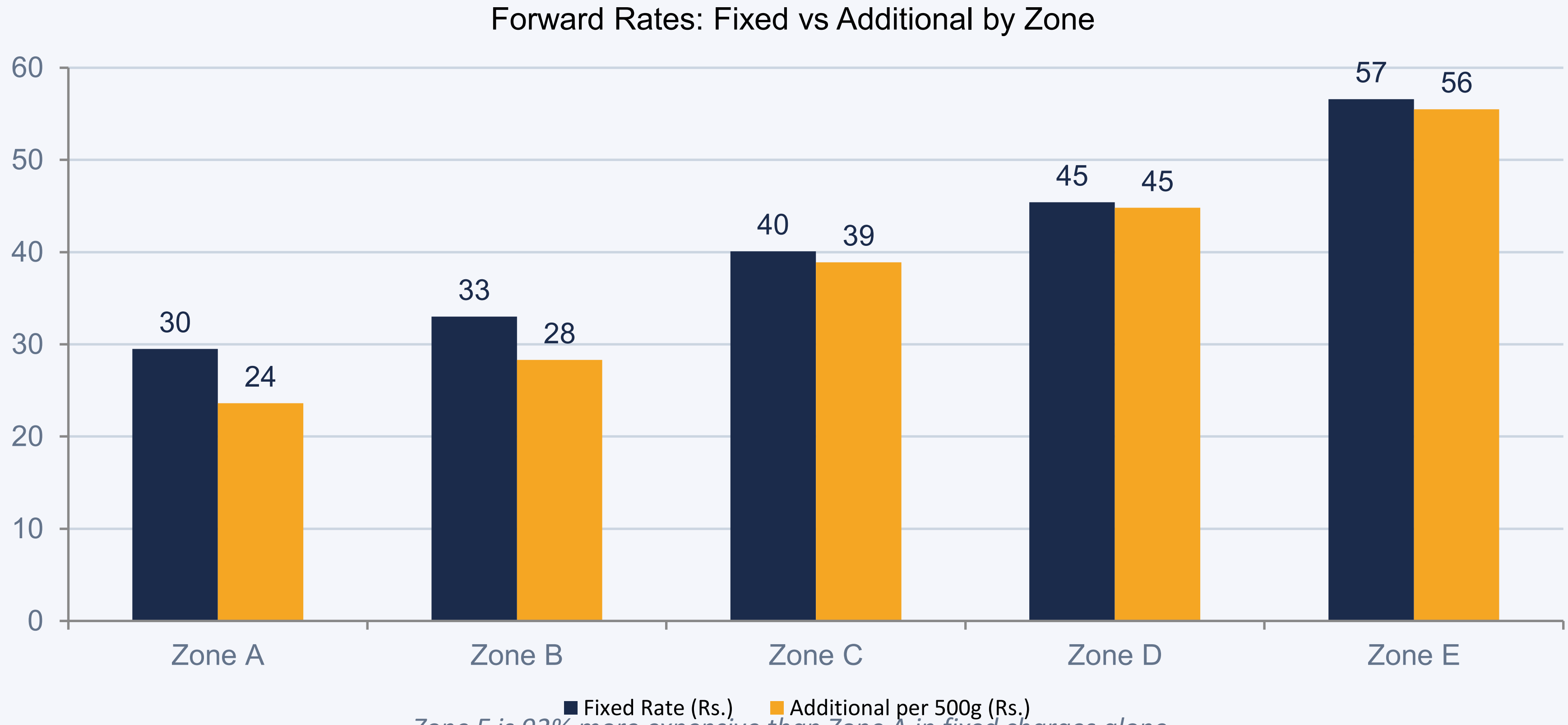
RTO Charges (Rs.) — Additional on top of forward charges

Zone	Fixed (Rs.)	Additional/500g (Rs.)
A	13.6	23.6
B	20.5	28.3
C	31.9	38.9
D	**41.3**	**44.8**
E	50.7	55.5

How Billing Works:

For weight ≤ 500g: Total = Fixed Charge
For weight > 500g: Total = Fixed Charge + [(weight – 500g) ÷ 500g] × Additional Charge
For RTO: Total = Forward Fixed + Forward Additional (if >500g) + RTO Fixed + RTO Additional (if >500g)

Forward Rate Comparison Across Zones

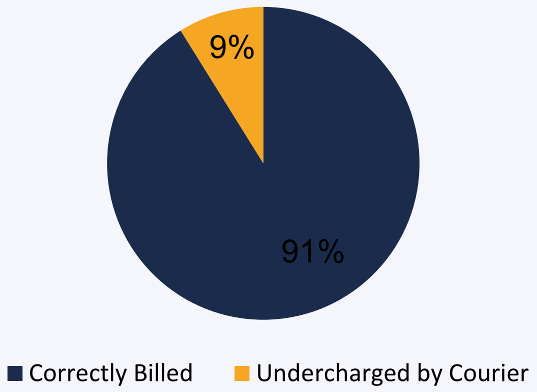


Zone E is 92% more expensive than Zone A in fixed charges alone.

Billing Discrepancy Analysis

₹13,648	₹13,718	₹70.20	11
Total Charged	Expected Total	Net Undercharged	Orders with Diff

Order Billing Accuracy



Findings
Out of 124 orders:

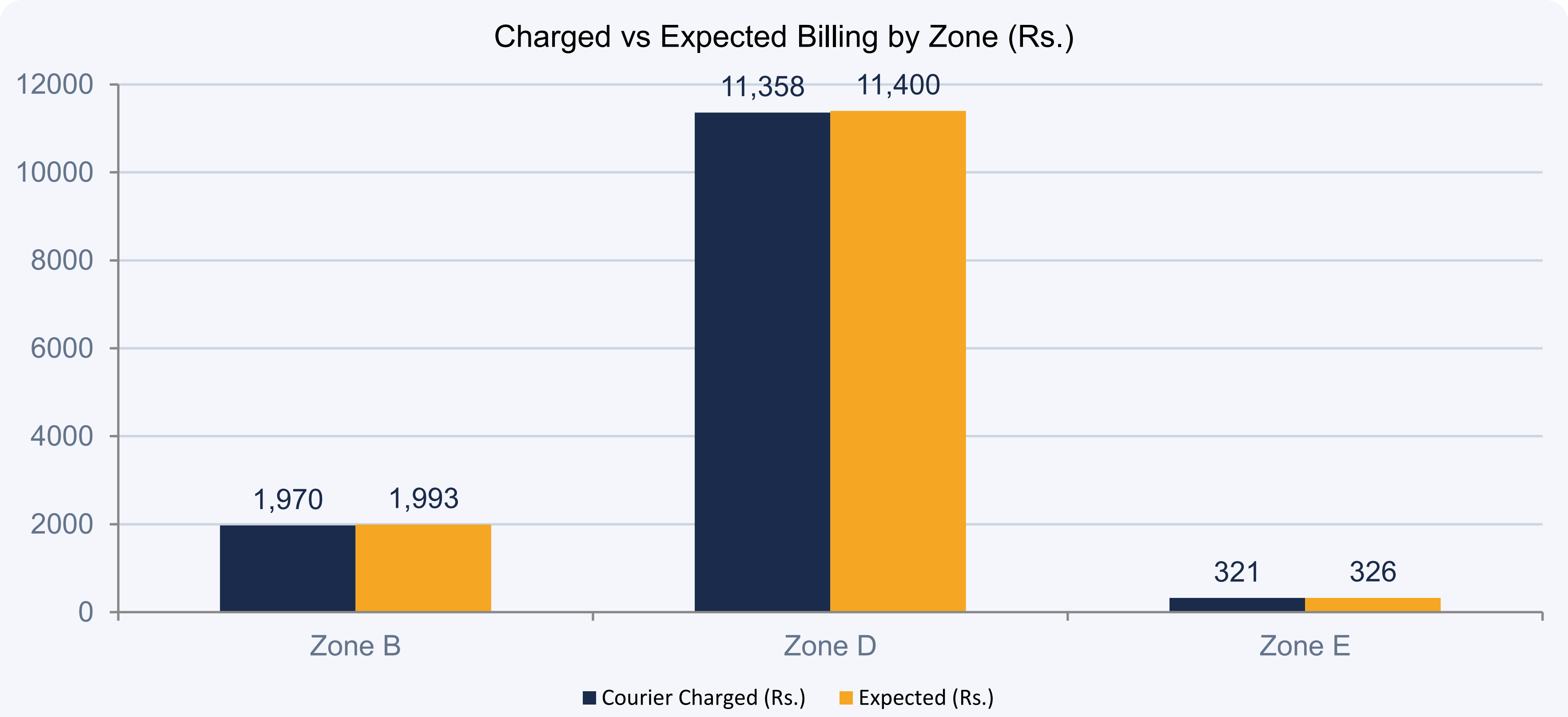
• 113 orders (91.1%) were billed exactly as expected.

• 11 orders (8.9%) were billed LESS than expected — Company X was undercharged by Rs. 70.20 total.

• 0 orders were overcharged.

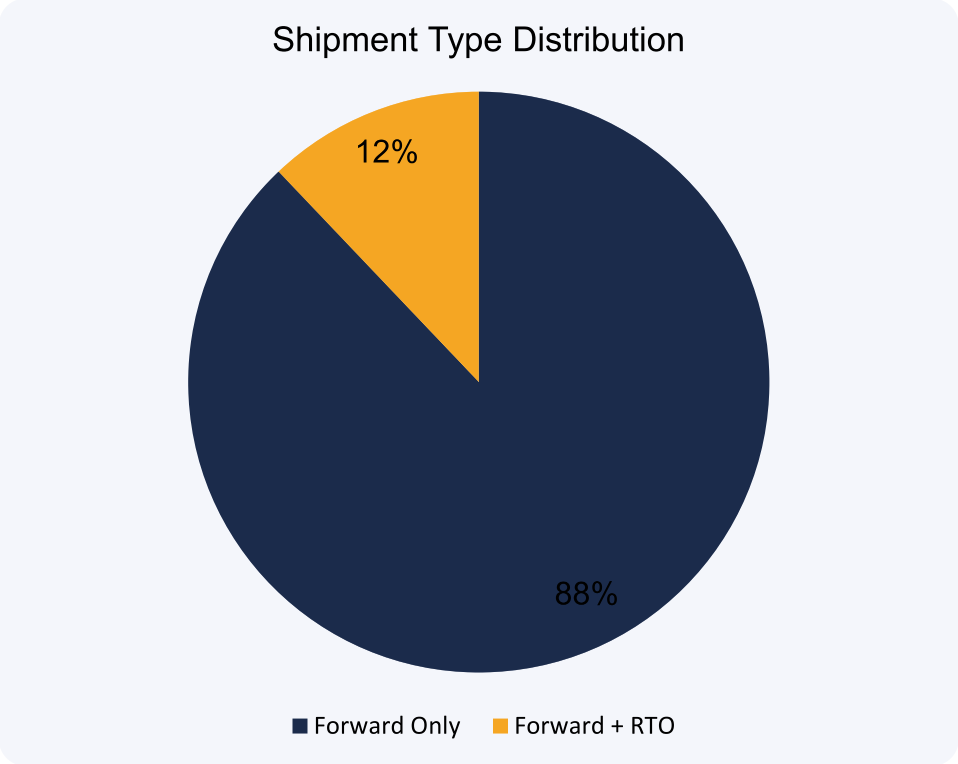
Overall, billing is accurate with a slight undercharge benefiting Company X.

Billing Discrepancy by Zone



In all zones, the courier billed slightly less than the expected amount based on our rate card calculations.

Shipment Type Analysis



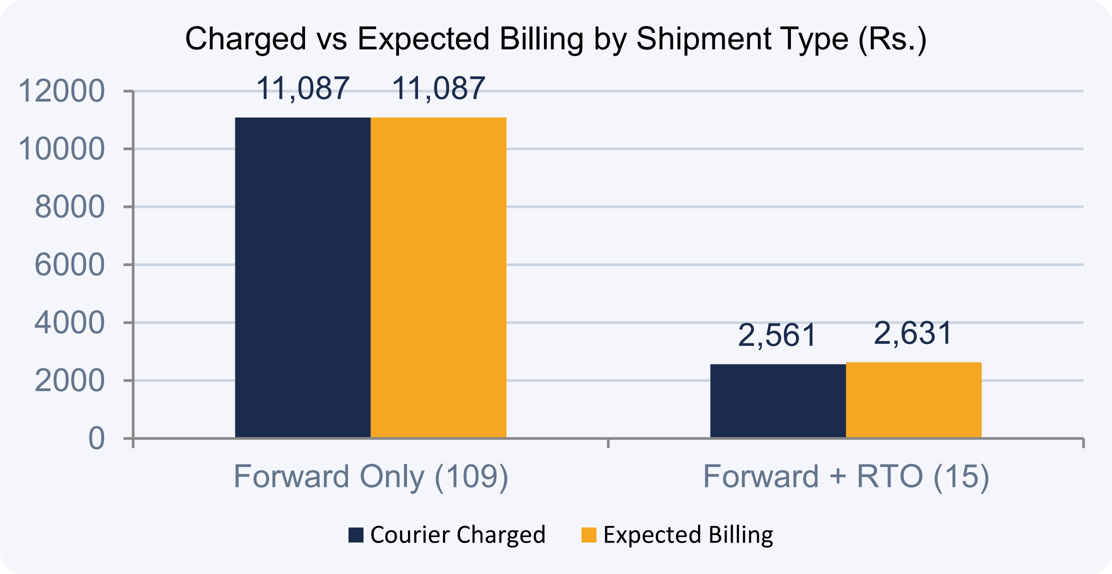
RTO Impact

12.1%	RTO Rate — 15 out of 124 orders returned
₹2,561	Total RTO Billing Costs
₹170.7	Average RTO+Forward Cost per Order
₹101.7	Average Forward-Only Cost per Order
68%	Higher cost for RTO vs Forward orders

RTO (Return to Origin) Deep Dive

What is RTO?

RTO (Return to Origin) occurs when an order cannot be delivered to the customer and is returned to the warehouse. It incurs double costs — the original forward shipping cost PLUS the return shipping cost.



Key Observations

Forward charges: Perfectly aligned — 0% discrepancy.

RTO charges: Courier billed ₹70.20 less than expected across 15 RTO orders.

All 15 RTO discrepancies were in Company X's favor.

Recommendation: Monitor RTO orders carefully and raise a credit note for any future overcharges.

Warehouse & Pincode Coverage

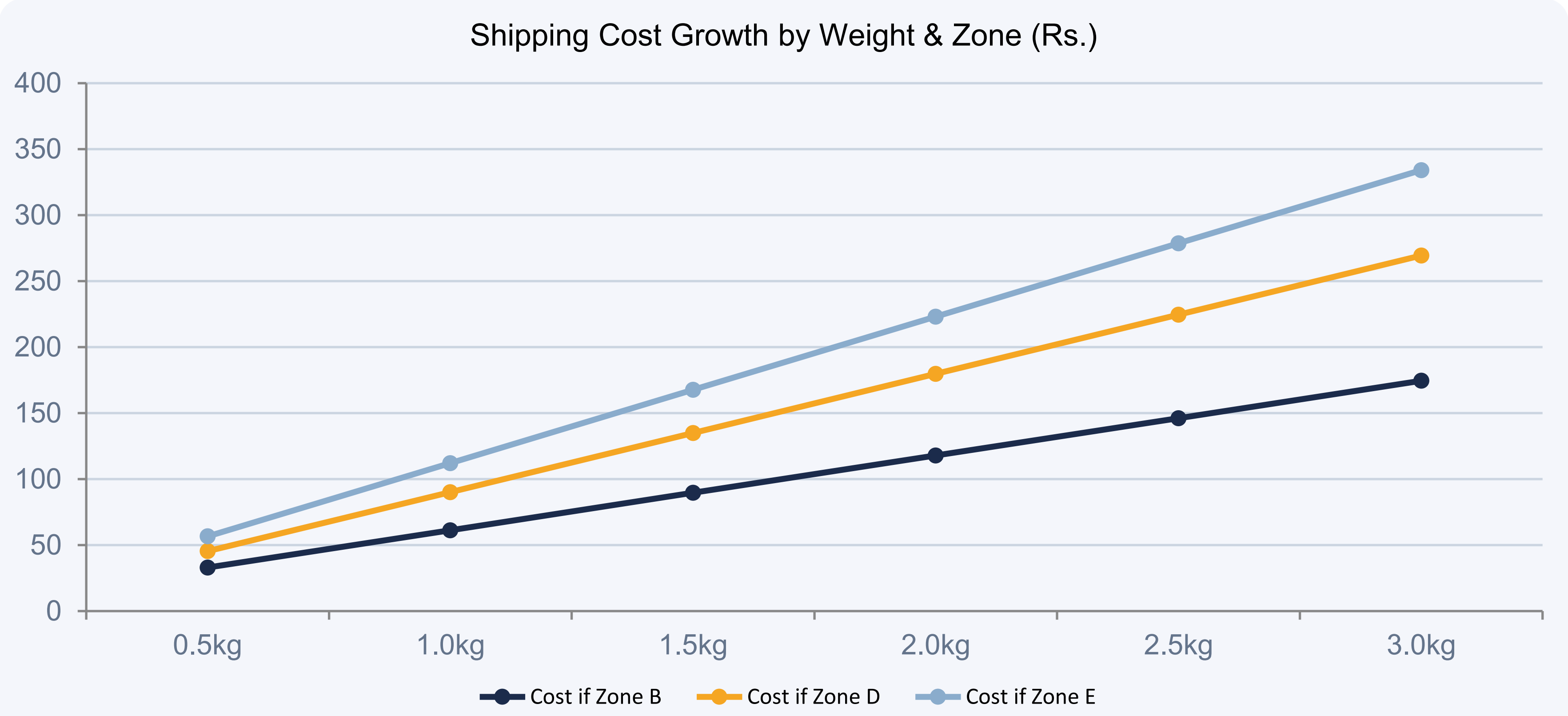
Single Warehouse Operation

All 124 shipments originated from a single warehouse with Pincode 121003 (Faridabad, Haryana). This centralized fulfillment model simplifies operations but concentrates supply chain risk.

Customer Pincode Spread

124	Unique Customer Pincodes Served
3	Zones Covered (B, D, E)
121003	Single Warehouse Pincode
79%	Shipments to Zone D

Shipping Cost Efficiency Analysis



Zone D costs 94% more than Zone B at the 2kg mark — highlighting the enormous impact of zone classification.

Supply Chain Process Flow

1	2	3	4	5	6
Order Placed	**SKU Picked**	**Weight Calculated**	**Zone Determined**	**Courier Handover**	**Billing Generated**
Customer places order on e-commerce platform	Warehouse team picks SKUs from inventory	Total weight computed from SKU master	Zone assigned via pincode mapping	Package handed to courier with AWB	Courier generates invoice based on rate card

Data Sources Used in This Analysis:

Order Report → SKU Master → Pincode Zones → Courier Rates → Courier Invoice | Cross-referencing all 5 datasets enables billing verification and zone audit.

Actionable Insights

01 Audit Zone Classifications

HIGH PRIORITY

60 shipments appear to be classified as Zone D when the pincode master suggests Zone B. At Rs. 45.40 vs Rs. 33.00 fixed charge, this alone represents ~Rs. 744 in potential overcharges per similar cycle.

02 Reduce RTO Rate

MEDIUM PRIORITY

At 12.1%, the RTO rate is significant. Each RTO shipment costs 68% more than a successful delivery. Improving delivery address verification and customer communication can reduce RTO losses.

03 Leverage Weight Consolidation

MEDIUM PRIORITY

With 18.5% of orders under 500g and many 2-3 item orders, there may be opportunities to bundle smaller items into single shipments to reduce per-unit shipping costs.

04 Automate Billing Reconciliation

QUICK WIN

Build a monthly automated reconciliation pipeline that cross-references all 5 data sources. This project proves that 8.9% of invoices had discrepancies — automation will ensure systematic catching.

Recommendations

Zone Audit

Raise a formal dispute with the courier over zone misclassification for 60+ shipments. Request a rate credit based on Zone B pricing.

RTO Reduction

Implement pre-delivery customer confirmation calls/SMS. Target reducing RTO rate from 12.1% to below 8%.

Weight Optimization

Explore packaging redesigns for top-selling lightweight SKUs to stay within the 500g billing slab.

Billing Automation

Deploy a monthly reconciliation script (Python/Excel) to auto-flag billing discrepancies > Rs. 5 per order.

Warehouse Expansion

Consider a second fulfilment centre to reduce Zone D/E shipments and lower logistics costs.

SKU Rationalization

Top 10 SKUs account for majority of volume — prioritize inventory buffers for these high-velocity items.

Summary of Key Findings

Billing Accuracy:	91.1% of orders (113 of 124) were billed correctly by the courier. No overcharging was found — the courier actually undercharged by Rs. 70.20 in total.
Zone Discrepancy:	Significant mismatch between invoice zones and pincode master: 60 shipments appear wrongly classified as Zone D (higher cost) vs Zone B (lower cost).
RTO Impact:	12.1% RTO rate drives 68% higher shipping costs per returned order. RTO charges totaled Rs. 2,561 against Rs. 11,087 in forward charges.
Weight Profile:	52.4% of shipments fall in the 500g–1kg bracket. Average SKU weight is 219g, which means most orders are lightweight.
Top SKUs:	Three SKUs — 8904223818850, 8904223818706, and 8904223819024 — account for the highest order volumes and drive the most shipping activity.
Single Warehouse:	All orders ship from one warehouse (Pincode 121003, Faridabad). Zone coverage is limited to B, D, and E — no Zone A or C shipments.

Methodology

1

Data Collection

Gathered 5 raw Excel files: Order Report, SKU Master, Pincode Zones, Courier Rates, and Courier Invoice. Verified data integrity and checked for nulls.

2

Data Merging

Joined Order Report with SKU Master on SKU field to calculate actual product weight per order line. Aggregated by Order ID.

3

Expected Billing

Applied the courier rate card formula (fixed + additional per 500g slab) for each order based on its zone, weight, and shipment type.

4

Discrepancy Analysis

Compared calculated expected billing with the courier invoice amount for each of the 124 orders. Flagged all differences.

5

Zone Verification

Cross-referenced the invoice zone against the pincode master zone for each order to identify misclassifications.

6

Insights & Reporting

Synthesized findings into visualizations and actionable recommendations for management review.

Conclusion

This analysis of Company X's supply chain and courier billing operations reveals a largely accurate billing relationship with the courier, but uncovers critical zone misclassification patterns that represent a significant cost risk.

The key takeaway is that while the numerical billing calculations are accurate (91.1% correct), the foundation of those calculations — zone assignment — appears flawed for a large proportion of shipments.

With the right reconciliation processes, audit mechanisms, and strategic decisions around warehouse expansion, Company X can significantly optimize its logistics costs and improve supply chain visibility.

Shivam Yadav | Data Science & Analytics | April 2026

Company X Supply Chain Analysis Project